

QUICK REFERENCE CARD - Engineering Economics Midterm

Print & Keep for Last-Minute Review!

TOP 10 CRITICAL FORMULAS (Memorize!)

1. Breakeven Quantity

$$BEQ = \frac{FC}{s - v}$$

- **FC** = Fixed Cost - **s** = Selling price per unit - **v** = Variable cost per unit - **Use:** “Find breakeven point” (90% of cost problems)

2. PED Midpoint Method

$$PED = \frac{|Q_2 - Q_1|}{|P_2 - P_1|} \times \frac{(P_2 + P_1)/2}{(Q_2 + Q_1)/2}$$

- **Use:** Given two (P, Q) points - **Always positive**, use midpoint as base

3. Profit

$$\text{Profit} = Q(s - v) - FC$$

- **Q** = Quantity - **Use:** Profitability at any output level

4. Total Revenue

$$TR = P \times Q$$

- **Use:** Revenue calculations, with elasticity

5. P/V Ratio

$$\text{P/V Ratio} = \frac{s - v}{s} \times 100$$

- **Use:** Contribution per dollar/rupee of sales - Alternative: Profit = (Sales $\times$ P/V) - FC

6. Income Elasticity (YED)

$$YED = \frac{\Delta Q_d / Q_d}{\Delta Y / Y}$$

- >1 = Luxury | $0-1$ = Necessity | <0 = Inferior - **Use:** “Classify good”

7. Cross-Price Elasticity (XED)

$$XED = \frac{\Delta Q_{d,A}/Q_{d,A}}{\Delta P_B/P_B}$$

- >0 = Substitutes | <0 = Complements - **Use:** “Relationship between goods”

8. Equilibrium

$$Q_d = Q_s$$

- **Use:** Set equal, solve for P

9. Marginal Cost

$$MC = \frac{\Delta TC}{\Delta Q}$$

- **Use:** Cost of one more unit

10. Marginal Revenue

$$MR = \frac{\Delta TR}{\Delta Q}$$

- **Use:** Revenue from one more sale

TOTAL REVENUE RULES (CRITICAL!)

Elasticity	Price $\uparrow$	Price $\downarrow$	Strategy
Inelastic (PED < 1)	TR $\uparrow$	TR $\downarrow$	RAISE price
Elastic (PED > 1)	TR $\downarrow$	TR $\uparrow$	LOWER price
Unit Elastic (PED = 1)	TR same	TR same	Price doesn't help

Use: “Should we increase price to increase revenue?”

ELASTICITY QUICK GUIDE

PED (Price Elasticity of Demand)

Value	Type	Demand Curve	Example
0	Perfectly Inelastic	Vertical	Life-saving drugs

Value	Type	Demand Curve	Example
<1	Inelastic	Steep	Necessities (salt)
$=1$	Unit Elastic	Medium	-
>1	Elastic	Flat	Luxuries (cruises)
∞	Perfectly Elastic	Horizontal	Perfect competition

YED (Income Elasticity)

Value	Type	Example
>1	Luxury (income elastic)	Sports cars
$0 \text{ to } 1$	Necessity (income inelastic)	Food, fuel
<0	Inferior good	Low-quality items

XED (Cross-Price Elasticity)

Value	Relationship	Example
>0	Substitutes	Tea & Coffee
<0	Complements	Car & Gas
$=0$	Unrelated	-

DECISION FLOWCHART

“Find breakeven”

→ $BEQ = FC / (s - v)$

“Two (P,Q) points”

→ PED Midpoint Method

“Profit at Q units”

→ $\text{Profit} = Q(s - v) - FC$

“Should raise price?”

→ Calculate PED → Check TR rule

“Substitutes or complements?”

→ XED (positive = sub, negative = comp)

“Luxury or necessity?”

→ **YED** (>1 = luxury, $0-1$ = necessity)

SAMPLE CALCULATIONS

Example 1: Breakeven

Given: $s = \$50$, $v = \$30$, $FC = \$500$
 $BEQ = 500/(50-30) = 500/20 = 25$ units

Example 2: PED Midpoint

Given: $P = \$10$, $Q = 100 \rightarrow P = \12 , $Q = 80$

Midpoints: $P=11$, $Q=90$

$$\% \Delta P = |12-10|/11 = 2/11 = 18.2\%$$

$$\% \Delta Q = |80-100|/90 = 20/90 = 22.2\%$$

$$PED = 22.2/18.2 = 1.22 \text{ (Elastic)}$$

Example 3: YED

Given: $Y = 500$, $Q = 20 \rightarrow Y = 600$, $Q = 25$

$$\begin{aligned} YED &= (5/20) / (100/500) \\ &= 0.25 / 0.20 = 1.25 \end{aligned}$$

$YED > 1 \rightarrow$ Luxury

Example 4: Profit

Given: $s = \$50$, $v = \$30$, $FC = \$500$, $Q = 30$

$$\begin{aligned} \text{Profit} &= 30(50-30) - 500 \\ &= 30(20) - 500 \\ &= 600 - 500 = \$100 \end{aligned}$$

KEY VARIABLES

Symbol	Full Form	Unit
BEQ	Breakeven Quantity	units
FC	Fixed Cost	Rs., \$

Symbol	Full Form	Unit
s	Selling price per unit	Rs./unit
v	Variable cost per unit	Rs./unit
Q	Quantity	units
P	Price	Rs./unit
Qd	Quantity Demanded	units
Qs	Quantity Supplied	units
TC	Total Cost	Rs., \$
TR	Total Revenue	Rs., \$
MC	Marginal Cost	Rs./unit
MR	Marginal Revenue	Rs./unit
Y	Income	Rs., \$
Δ	Change (Delta)	-

COMMON TRAPS TO AVOID

DON'T:

1. Use **FC/v** for BEQ (WRONG!)
Use **FC/(s-v)**
2. Use starting value as base for %
Use **midpoint** for PED
3. Drop sign from XED
Keep sign (tells substitutes vs complements)
4. Report PED as negative
Always take **absolute value**
5. Forget units in final answer
Always write **units**, \$, %, etc.

COST FORMULAS

Total Cost

$$TC = TVC + TFC = (v \times Q) + FC$$

Contribution

$$\text{Contribution} = s - v$$

Margin of Safety

$$\text{MOS} = \text{Actual Sales} - \text{BE Sales}$$

$$\text{MOS}\% = \frac{\text{Actual} - \text{BE}}{\text{Actual}} \times 100$$

Profit using P/V

$$\text{Profit} = (\text{Sales} \times \text{P/V Ratio}) - FC$$

Required Sales for Target Profit

$$\text{Required Sales} = \frac{FC + \text{Desired Profit}}{\text{P/V Ratio}}$$

MARKET EQUILIBRIUM

Equilibrium

$$Q_d = Q_s$$

Shortage (Excess Demand)

$$Q_d > Q_s \Rightarrow \text{Price} \uparrow$$

Surplus (Excess Supply)

$$Q_s > Q_d \Rightarrow \text{Price} \downarrow$$

PRE-EXAM CHECKLIST

- ☐ Can calculate BEQ without looking
 - ☐ Know midpoint PED by heart
 - ☐ Understand TR-elasticity rules
 - ☐ Can interpret XED sign
 - ☐ Know YED classification
 - ☐ Memorized s, v, FC, Q meanings
 - ☐ Can solve breakeven table
 - ☐ Know movement vs shift
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EXAM STRATEGY

1. Read ALL questions first (5 min)

Identify easy ones, mark priorities

2. Do in order:

1. **Easy ones** (quick points)
2. **Breakeven/PED** (most important)
3. **Profit calculations**
4. **Harder ones**

3. Always show:

Formula first

Variable definitions

Substitution

Calculation steps

Final answer with units

4. Time per question:

- Simple calculation: 3-5 min
 - Breakeven table: 8-10 min
 - Elasticity with interpretation: 5-7 min
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CONFIDENCE CHECKLIST

I know the top 10 formulas

I can do midpoint PED in my sleep

I understand BEQ concept

I know when to use each formula

I can interpret elasticity values

I understand TR rules

FINAL REMINDERS

1. **BEQ is king** - appears in 90% of problems
 2. **Midpoint method** - default for PED
 3. **Show your work** - partial credit exists
 4. **Label everything** - s, v, FC, Q, etc.
 5. **Check units** - \$, Rs., %, units
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YOU'RE READY!

Trust your preparation. Read carefully. Show your work.

Good Luck!
